

Predicting Student Exam Performance Using Supervised Machine Learning

Osamuyimen Osunde

Student ID: 20093878

B9DA109 — Machine Learning and Pattern Recognition

Dublin Business School

6,607

Rows

14

Features

3

ML Models

84.49%

Best R^2

Problem, Dataset & Algorithm Selection

Real-World Problem

Can we predict a student's exam score from measurable factors like study time, attendance, and support systems? An accurate model lets schools flag at-risk students early enough to intervene.

Dataset

Source: Kaggle — Student Performance Factors
6,607 rows × 20 cols → 14 features after removal

Missing: Teacher_Quality & Parental_Education_Level imputed

Target: Exam_Score (continuous → regression)

Algorithm Selection

1. Decision Tree Regressor

Interpretable baseline. Tree depth controls complexity — splits the feature space via threshold rules readable by non-technical stakeholders.

2. Random Forest Regressor

Bagging ensemble — builds many trees and averages predictions. Reduces variance and captures feature interactions naturally.

3. Support Vector Regressor

Kernel-based method. Maps features to higher-dimensional space; robust to outlier students via epsilon-insensitive loss.

Three algorithms spanning interpretable → ensemble → kernel complexity.

Data Preparation & Feature Selection

Feature Removal

5 columns dropped: Gender, School_Type, Physical_Activity, Sleep_Hours, Distance_from_Home. No consistent directional link to exam score. Final: 14 features.

Data Encoding

10 categorical columns mapped to ordered integers — e.g. Low=0, Medium=1, High=2. Preserves natural ranking that one-hot encoding would destroy.

Train / Test Split

90/10 split (random_state=100): 5,946 train / 661 test. Larger training share chosen given the moderate dataset size of 6,607 rows.

Data Clean/Missing Value Imputation

Teacher_Quality (78 missing) and Parental_Education_Level (90 missing) filled with their column mode. Avoids dropping rows; minimal distortion given small missing %.

Feature Scaling

StandardScaler applied to all 14 features. Essential for SVR (distance-based). Prevents large-scale features dominating small-scale ones.

Data Balancing

Regression predicts a continuous value — class imbalance does not apply. Step explicitly skipped as noted in the notebook.

Model Development, Optimisation & Comparison

Decision Tree

Test R^2

54.68%

CV R^2 (tuned)

50.30%

max_depth=5

Random Forest

Test R^2

78.86%

CV R^2 (tuned)

63.72%

n_estimators=200

SVR ★ Best

Test R^2

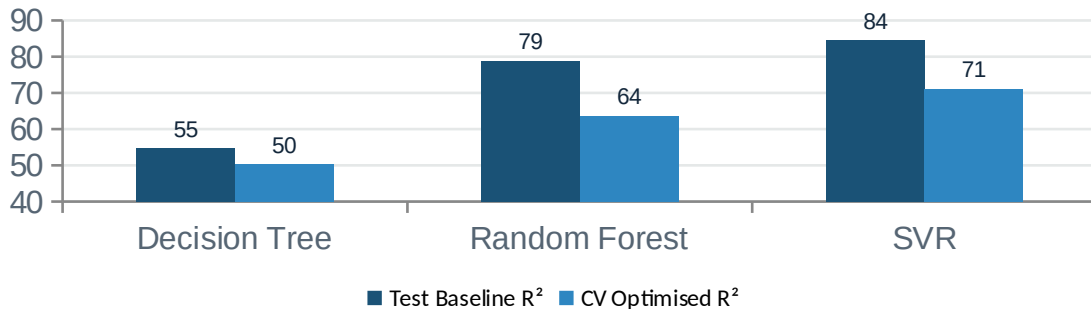
84.49%

CV R^2 (tuned)

71.17%

C=100, linear

R^2 Score Comparison (%)



Best Model

SVR
84.49% R^2

poly kernel | test set

Conclusion, Recommendation & References

54→85%

R^2 Range

Models ranged from 54.68% (DT) to 84.49% (SVR) — a 30-point spread confirming model choice matters.

SVR Wins

Best Model

SVR leads on every metric. Linear kernel selected by GridSearchCV shows relationships are approximately linear once scaled.

CV Gap

Key Insight

All test scores exceed CV scores — the 90/10 split produced an easier test set. SVR's lead holds on both measures.

Conclusion

Three supervised regression models were applied to predict student exam scores from 14 features. After domain-based feature removal, mode imputation, ordinal encoding, and StandardScaler normalisation, SVR emerged as the strongest model at 84.49% test R^2 . All 14 inputs are routinely collected by schools — the model could function as a mid-term early-warning system to flag at-risk students and trigger targeted support before results are finalised.

References

Abdel-Hak, A. (2026) Machine Learning and Pattern Recognition. | Bergstra & Bengio (2012) Random search for hyper-parameter optimisation, JMLR 13. | Breiman (2001) Random Forests, ML 45(1). | Cortes & Vapnik (1995) Support-vector networks, ML 20(3). | Kaggle (2024) Student Performance Factors [dataset]. | Pedregosa et al. (2011) Scikit-learn, JMLR 12.